

**23ARE301: INTRODUCTION TO DATA SCIENCE**  
**Project Review 0**

# Predicting Grasp Stability in Robotic Hands

*A study of joint kinematics from a simulated robotic hand, used to predict whether a grasp will hold or fail.*

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## TEAM MEMBERS

**A Gowtham Satya Sri Ram**

CB.EN.UARE24002

**K Kranthi Koushik Reddy**

CB.EN.UARE24017

**Kishor Kumar U**

CB.EN.UARE24019

**M Shanmukha Vinayak**

CB.EN.UARE24023

**P Lohith**

CB.EN.UARE24034

# Dataset & Source

***DATASET:** Grasping Dataset — Shadow Robot Smart Grasping Sandbox*

## Overview

- Public benchmark dataset for robotic grasp stability prediction.
- Collected using the Shadow Robot Smart Grasping Sandbox in ROS Gazebo.
- Suitable for regression and classification tasks in robotics and machine learning.

## Feature Set

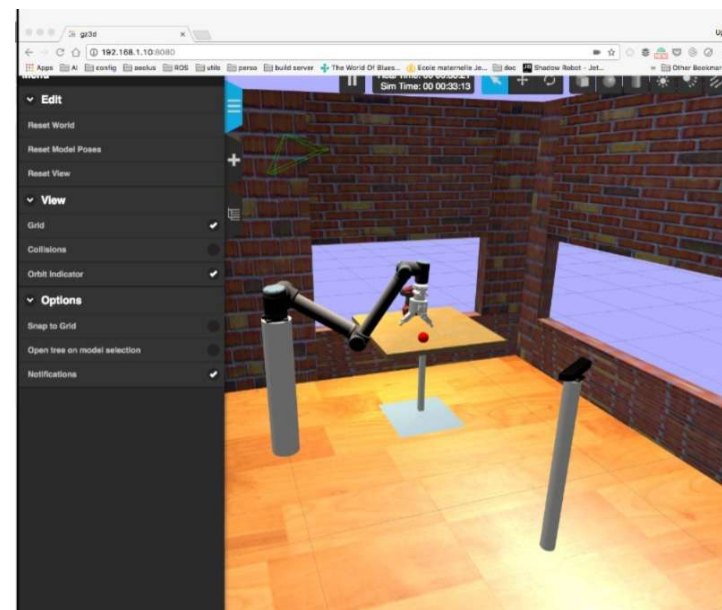
Each grasp attempt records:

- Joint Position (Angle)
- Joint Velocity
- Joint Effort (Torque)

for every joint of the robotic hand.

## Target Variable:

Grasp Robustness Score: Quantifies the displacement of the grasped object after an external disturbance.



Grasp attempts recorded

**~54,000**

Measurements per experiment

**29**

Fingers on the robotic hand

**3 (3-DOF each)**

# Methodology - Preparing the Data

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## 1. Data Cleaning

- Check all 54,000 grasp records for missing or corrupted joint readings.
- Handle incomplete rows drop them or impute using column-wise mean/median.
- Detect and remove outliers (e.g. impossible joint angle or torque values)

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## 2. Data Preprocessing

- Normalize/scale joint angle, velocity, and torque features to a common range.
- Ensure consistent scaling across all three fingers of the hand.
- Separate the 29 measurements into input features (X) and target variable (y)

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## 3. Feature Labeling

- Study the distribution of the continuous robustness score (target variable)
- Define a threshold to convert it into two classes: Stable / Unstable
- Encode the new class label for use in classification

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## 4. Data Visualization

- Histogram of robustness scores to check class balance
  - Scatter plots of key joint features vs. robustness
  - Correlation heatmap across all 29 features to identify redundant variables
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# Methodology - Model Building & Validation

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## 5. Model Development

- Split the processed dataset into training and testing subsets.
  - Develop a Linear Regression model to predict the continuous grasp robustness score.
  - Develop a Logistic Regression model to classify grasps as Stable or Unstable.
  - Compare model performance using appropriate evaluation metrics.
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## 6. Model Evaluation

### • Regression

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Coefficient of Determination ( $R^2$ )

### • Classification

- Accuracy
  - Precision
  - Recall
  - F1-Score
  - Confusion Matrix
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## 8. Expected Outcome

- Predict the grasp robustness score from joint angle, velocity, and torque measurements.
- Classify grasp attempts as Stable or Unstable based on the predicted robustness.
- Enable early identification of potentially unstable grasps, supporting more reliable robotic manipulation.

# References

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## 1. DATASET

Cupcic, U. Grasping Dataset. Kaggle.  
[kaggle.com/datasets/ugocupcic/grasping-dataset](https://kaggle.com/datasets/ugocupcic/grasping-dataset)

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## 2. GITHUB REPOSITORY

Shadow Robot Company. smart\_grasping\_sandbox. GitHub repository.  
[github.com/shadow-robot/smart\\_grasping\\_sandbox](https://github.com/shadow-robot/smart_grasping_sandbox)

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## 3. BACKGROUND READING

Cupcic, U. How I Taught My Robot to Realize How Bad It Was at Holding Things. Shadow Robot Blog.  
[shadowrobot.com](https://shadowrobot.com)